

HARSH SAW

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SUMMARY

Production systems engineer with 2+ years of internship experience owning CI/CD pipelines, release automation, and cloud deployment infrastructure. Unified a polyglot monorepo under Bazel at OmMuse, operated Kubernetes and Prometheus/Grafana stacks at Bwisher, and provisioned full AWS infrastructure with Terraform at Jythu — delivering zero-downtime releases and observable systems serving tens of thousands of users.

EDUCATION

Bachelor of Computer Information Systems
Okanagan College

Expected Dec 2026
Kelowna, BC, Canada

EXPERIENCE

- OmMuse** Seattle, US (Remote)
Software Engineer Intern *Nov 2025 – Present*
- Shipped features across a polyglot monorepo (**Go**, **TypeScript/React**, **Python**) unified by **Bazel**, owning full deployment infrastructure — AWS (S3, SQS, ECR), **Docker**, **Pulumi IaC**, Envoy proxy, and **GitHub Actions CI/CD**.
 - Built HubSpot CRM sync for a **12K+** user platform in **Go** — async batch worker (sync.Map dedup, 30s flush, 100-user chunks) across 8 API trigger points; platform manages 18K+ tracks and 67 GB of audio.
 - Engineered a cross-platform desktop uploader (Electron) with folder-sync engine (chokidar + SQLite WAL), parallelized **S3 multipart uploads**, OAuth 2.0 PKCE, and typed IPC bridge over 70+ **gRPC** methods.
 - Architecting AI-powered semantic search via RAG pipeline (MongoDB Vector Store, AWS Bedrock) enabling natural language queries across projects, tracks, and collaborators.
- MoreThinks Solutions Ltd.** Burnaby, BC
Full-Stack Developer Intern *Jun 2025 – Sept 2025*
- Delivered full-stack features in a monorepo with **CI/CD via AWS EKS** and API Gateway; engineered token generation and **vector embedding pipelines** consumed by ML recommendation services.
- Bwisher Ltd.** Remote
Software Engineer Intern *Dec 2024 – Jun 2025*
- Owned end-to-end deployment with **Jenkins**, **Kubernetes**, **Prometheus**, and **Grafana**; shipped via feature flags enabling safe per-tenant rollouts with instant kill-switch control.
 - Architected a multi-channel platform using **Apache Kafka** + BullMQ, orchestrating SMS/email/WhatsApp campaigns across **30K+** users while feeding real-time CRM and analytics pipelines.
 - Designed idempotent **NestJS** + **FastAPI** microservices for payment/credit workflows with **retry and exponential backoff**, eliminating duplicate transactions; built demand forecasting services for ML-driven offer segmentation.
- Jythu Ltd.** Remote
Full-Stack Developer Intern *Jan 2024 – Dec 2024*
- Provisioned full AWS infrastructure with **Terraform** + **GitHub Actions CI/CD** for two production apps serving 2K+ learners; built device-bound anti-piracy auth and FFmpeg HLS streaming pipeline (SQS/Lambda/SNS) cutting seek time **30s → 1.5s**, bandwidth **–70%**.

PROJECTS

- CodeExpo — Browser-Based IDE & Sandbox** | *React, Node.js, Docker, Nginx, xterm.js, WebSocket* [Demo](#) | [GitHub](#)
- Full-stack browser IDE with live terminal (**xterm.js** over WebSocket), real-time code execution, and Socket.IO file sync; **Docker** multi-stage containerized sandboxes with Nginx reverse proxy and isolated per-session execution contexts.
- Document Intelligence Platform (RAG)** | *FastAPI, LangChain, Qdrant, Groq, Gemini, AWS ECS Fargate* [Demo](#) | [GitHub](#)
- Multi-tenant RAG platform with session-isolated **Qdrant** indexes; resilient LLM failover (Groq→Gemini) via LangChain; deployed to **AWS ECS Fargate** with **GitHub Actions CI/CD** and SHA-256 idempotent ingestion.

TECHNICAL SKILLS

Languages: Python, Go (Golang), TypeScript, JavaScript (ES6+), Bash
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Pulumi, **Bazel**, Jenkins, GitHub Actions
Backend & Messaging: NestJS, FastAPI, Express, gRPC/Protobuf, Apache Kafka, BullMQ, RabbitMQ
Observability: Prometheus, Grafana, LangSmith
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite
Frontend: React, Next.js, React Native, Qwik.js, Electron
AI & GenAI: LangChain, RAG, FAISS, Qdrant, Pinecone, AstraDB, AWS Bedrock

ACHIEVEMENTS

- **500 S&P 500** stock prediction pipeline: XGBoost across 26 intraday horizons on **4x NVIDIA H100 GPUs** (DRAC) with warm-start retraining every 15 minutes. Top **2%** freelancer on Freelancer.com (5-star, 25+ production projects). Won inter-university hackathon.